

Institute of Engineering and Technology, Lucknow
B. Tech Class Test-I (2024-25)

Branch: Mechanical Engineering

Semester: IVth

Subject Name: Practices and Ethical Issues in Research

Subject Code: INC 405

Time: 1 hour

Max. Marks: 20

Note: Attempt all Questions. All Question carry equal marks.

Q. No.	Question	Marks	CO	BL
Q1.	Describe the term Ethics. Explain the three philosophical approaches of ethics in brief.	(5)	1	L2
Q2.	What is research? Explain the characteristics and objective of research.	(5)	1	L2
Q3.	Explain the term: (a) Research Misconduct (b) Fabrication and Falsification	(5)	2	L2
Q4.	What do you understand by plagiarism? What are the parameters of ethics of using text from earlier work?	(5)	2	L1

Maximum Marks: 20

Time: 1 Hr

B.Tech - 4th Semester-2024-25 Makeup CT: Applied Thermodynamics (IME-401)

Department of Mechanical Engineering
Institute of Engineering and Technology, Lucknow

Note: Use of steam table is permitted.

Q. No.	Question	CO/BL level	Marks
1	The net power output of an ideal regenerative-reheat steam cycle is <u>80 MW</u>. Steam enters the h.p. turbine at 80 bar, 500° C and expands till it becomes saturated vapour. Some of the steam then goes to an open feedwater heater and the balance is reheated to 400° C, after which it expands in the l.p. turbine to 0.07 bar. Compute (a) the reheat pressure, (b) the steam flow rate to the h.p. turbine, and (c) the cycle efficiency. Neglect pump work. OR Propane is burned with 250% theoretical air, both are at 25 °C and 1 atm. Find the adiabatic flame temperature in the steady flow combustion chamber. Use enthalpy of formation of Propane, CO₂, and H₂O as (-)103850 kJ/kmol, (-)393250 kJ/kmol, (-)241820 kJ/kmol respectively. Specific heat of O₂, N₂, CO₂, and H₂O is 30 kJ/kmol-K, 29 kJ/kmol-K, 36 kJ/kmol-K, 33 kJ/kmol-K respectively.	CO-2 BL-3	10
2 (a)	Show that the height of chimney required for producing a draught equivalent to 15 mm of water cannot be less than 30, if the fuel gas temperature is 250° C, ambient temperature is 20° C and minimum 18 kg air per kg of fuel is required.	CO 3/ BL 3	5
(b)	A convergent-divergent nozzle has steam entering at 10 bar and 0.97 dry and leaving at 0.6 bar. For the steam mass flow rate of 0.2 kg/s determine the ratio of throat to exit diameter considering that 10% of overall isentropic heat drop is the frictional heat generated in divergent portion of nozzle.	CO3/ BL3	5

IET Lucknow

B.Tech IVth Semester, Branch: ME, CS(AI)

Mid sem Test-1

Sub-Computer system security(INC-401)

Note: Attempt any five questions. Each questions carry equal marks

Time: 1:00 Hr

Total Marks: 20

- Q1. What do you mean by Computer System Security? Explain different type of computer Security?
- Q2 What do you mean by Attacks? Differentiate between active and passive attacks with suitable examples.
- Q3. Explain buffer overflow. Discuss its impact on system security.
- Q4. Explain the risks posed by malicious software such as viruses, worms and spyware. How do they differ from each other in their behavior and impact?
- Q5. Explain the significance of Access Control in Unix/Linux system and discuss the role of permissions for files and directories.
- Q6. Define the following terms , (i) Message Integrity (ii) Message Availability

Department of Mechanical Engineering
Institute of Engineering and Technology, Lucknow
B.Tech - (ME) - 4th Semester-2024-25 CT-1: Applied Thermodynamics (IME-401)

Maximum Marks: 20

Time: 1 Hr

Note: Use of steam table is permitted.

Q. No.	Question	CO/BL level	Marks
1 (a)	(i) Draw p-v plot of rankine cycle with saturated steam at turbine outlet. What is the effect of regeneration on power output, cycle efficiency and steam rate. (ii) Why is one open feed water heater used in a steam power plant? what is it called?	CO-2 BL-2	3 +2
(b)	In a reheat cycle, the initial steam pressure and the maximum temperature are 150 bar and 550° C respectively. If the condenser pressure is 0.1 bar and the moisture at the condenser inlet is 5%, and assuming ideal processes, determine (a) the reheat pressure if the steam is reheated to 550 °C, (b) the cycle efficiency, and (c) the steam rate.	CO2/ BL3	5
2 (a)	Make comparison between fire tube and water tube boilers. Also discuss why high pressure boilers are generally water tube boiler.	CO 3/ BL 2	5
(b)	Air is expanded reversibly and adiabatically in a nozzle from 15 bar and 300 °C to a pressure of 3 bar. The inlet velocity of the nozzle is very small and the process occurs under steady flow conditions. Calculate the velocity at the throat and exit of the nozzle.	CO3/ BL3	5

Institute of Engineering and Technology, Lucknow

B. Tech Class Test-I (2024-25)

Branch: Mechanical Engineering

Semester: IVth

Subject Name: Practices and Ethical Issues in Research

Subject Code: INC 405

Time: 1 hour

Max. Marks: 20

Note: Attempt all Questions. All Questions carry equal marks.

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Q1.	Describe the term Ethics. Explain the three philosophical approaches of ethics in brief.	(5)	1	L2
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Q3.	Explain the term: (a) Research Misconduct (b) Fabrication and Falsification	(5)	2	L2
Q4.	What do you understand by plagiarism? What are the parameters of ethics of using text from earlier work?	(5)	2	L1

IET Lucknow

B.Tech IVth Semester, Branch: ME, CS(AI)
Mid Sem Test-2

Sub-Computer system security(INC-401)

Note: Attempt any five questions. Each questions carry equal marks

Time: 1:00Hr

Total Marks: 20

Q1. Discuss TCP session Hijacking include its mechanism and potential impact on security.

Q2. Explain the Bell-Lapadula model. Discuss its significance in maintaining data confidentiality.

Q3. What do you mean by Firewall? Differentiate between Intrusion Detection, and Intrusion Prevention Systems with suitable example.

Q4. (b) What do you mean by authentication and message Integrity in cryptography? How can you achieve these properties in symmetric cryptography?

Q5. Define digital signature. Describe the different step involved in creating digital signature

Q6. Define the following terms, (i) public Key Infrastructure(PKI) (ii) Electronic Payment System

Institute of Engineering and Technology, Lucknow

B. Tech Class Test-II (2024-25)

Branch: Mechanical Engineering

Subject Name: Practices and Ethical Issues in Research

Time: 1 hour

Semester: IVth

Subject Code: INC 405

Max. Marks: 20

Note: Attempt all Questions. All Question carry equal marks.

Q. No.	Question	Marks	CO	BL
Q1.	What do you understand by Bibliography? What are the different citation styles used for writing a research paper?	(5)	3	L1
Q2.	Describe the emerging technologies and tools that can be used to support the application of design thinking methods to scientific writing and research.	(5)	3	L2
Q3.	How can organizations and institutions use anti-plagiarism software to promote academic integrity among students and researchers?	(5)	4	L2
Q4.	What is the difference between secrecy and confidentiality? How are these concepts applied in different contexts?	(5)	4	L1

Institute of Engineering & Technology Lucknow
Department of Mechanical Engineering
B.Tech. Class Test-II (2024-25)

Branch: Mechanical Engineering
Subject Name: Manufacturing Processes
Time: 1 hour

Semester: 4th
Subject Code: IME-403
Max. Marks: 20

Note: Attempt all four Questions. All Question carry equal marks.

	Marks	CO	BL
Q. 1: Explain submerged arc welding with neat sketch. Or Compare TIG welding with MIG welding with its schematic diagram.	5	4	2
Q. 2: Explain fusion as it relates to welding operations. How will you classify the welding processes? Explain the features of neutral, reducing, and oxidizing flames.	5	4	2
Q. 3: What do you understand by grit, grade, bond and structure of a grinding wheel?	5	5	2
Q. 4: What is centerless grinding? Draw its schematic diagram. Differentiate between through feed and in-feed types.	5	5	2

Institute of Engineering and Technology, Lucknow
B.Tech. CT-2 (2024-25)

Branch: ME

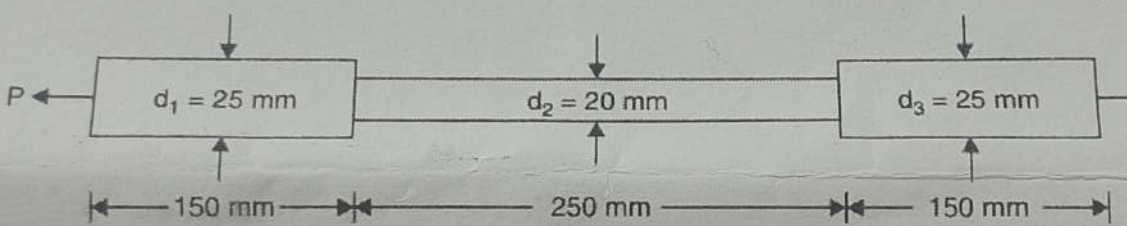
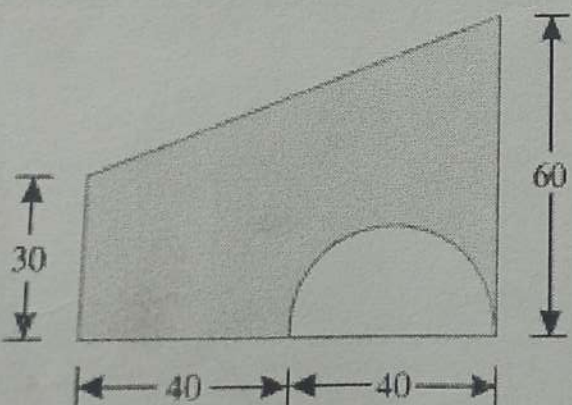
Semester: 4

Subject: Engineering Mechanics & Strength Of Materials Code: IME402

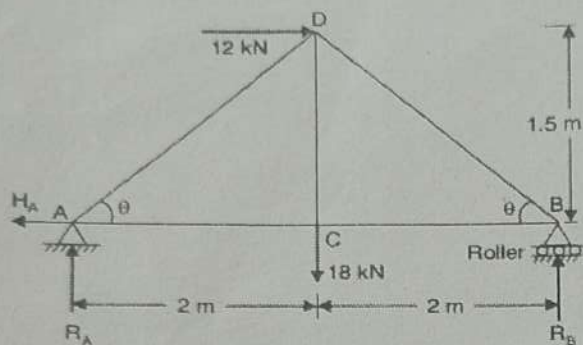
Time: 1 hrs

Max. Marks: 20

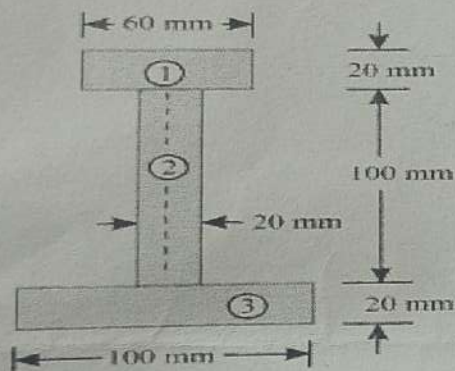
Note: Attempt all Questions. All Question carry equal marks.

Q. No.	Question	Marks	CO	BL
1 (A)	A block weighing 500 N just starts moving down a rough inclined plane when it is subjected to 200 N force acting up the inclined plane and it is at the point of moving up the plane when pulled up by a force of 300 N parallel to the plane. Find the inclination of the plane and the coefficient of friction between the inclined plane and the block.	5	1	K3
1 (B)	The bar shown in the figure is tested on the universal testing machine. It is observed that at a load of 40 kN the total extension of the bar is 0.280 mm. Determine the Young's modulus of the material.  OR A semicircular area is removed from a trapezium as shown in Fig (dimensions in mm) 	5	1	K3

2 (A) Determine the forces in the truss shown in Fig. which carries a horizontal load of 12 kN and a vertical load of 18 kN.

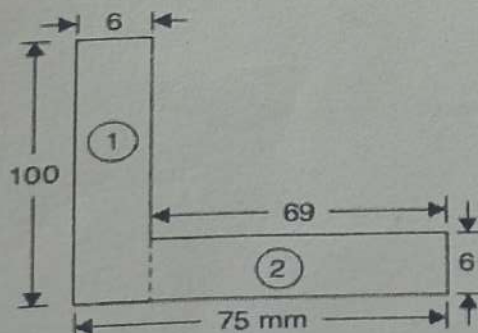


2 (B) An I-section is made up of three rectangles as shown in Fig.. Find the moment of inertia of the section about the horizontal axis passing through the centre of gravity of the section.



OR

Find the moment of inertia of ISA 100 × 75 × 6 about the centroidal XX and YY-axis, shown in Fig.



Department of Mechanical Engineering
Institute of Engineering and Technology, Lucknow
4th Semester-2024-25 CT-2: Applied Thermodynamics (IME-401)

Maximum Marks: 20

Time: 1 Hr

Note: Use of steam table is permitted.

Q. No.	Question	CO/BL level	Marks
1 (a)	Define the adiabatic flame temperature in combustion process. In a certain combustion process, Butane is burnt with 20% excess air. The combustion is complete one that is carried out at a total pressure of 3 bar. Determine the air fuel ratio and dew point temperature of the product.	CO 1/ BL 3	5
(b)	Compare 2-s and 4-s engine on the basis of mechanical efficiency. An air standard ideal diesel engine operates with a pressure and temperature at the start of the compression stroke of 1 atm. and 30 C. If the cutoff ratio and clearance volume are 2.5 and 0.1 m ³ respectively, determine pressure and temperature at all points and thermal efficiency .	CO 1/ BL 3	5
2 (a)	Determine nozzle angle, blade angles at inlet and exit for a single stage impulse steam turbine developing 132 kW, speed of 3340 rpm, mean rotor diameter of 1 m, steam flow rate of 2 kg/s, blade velocity coefficient of 0.9, steam leaving nozzle at 400 m/s and steam discharge axially from turbine. Solve using velocity diagram.	CO 4/ BL 3	5
(b)	Draw the pressure, velocity and specific volume variation along the axis of the Pressure-velocity compounded steam turbine.	CO 4/ BL 2	5

IVE 401

Universal Human Values

CT-2

29-04-2024

1. What are the right sources for continuous happiness? Elaborate with example.
2. 'I am responsible for the nurture, protection and right utilization of the Body'. Elaborate this statement
3. What are the key differences between happiness from material consumption and happiness from feelings?
4. How any why, ego in human behavior leads to depression?
5. Differentiate between Self and Body on the basis of Needs and Activities.